

Worksheet 1 – Basics of KVL and KCL

Total points – 20 (Q7&8 carry 4 points each. Others 2 points each)

Attempt all problems

Mention all node/loop variables on the circuit diagrams

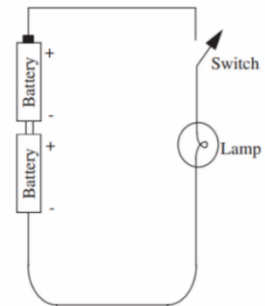
Mention all methods/equations used for each problem

Mention all assumptions (if any) made while solving

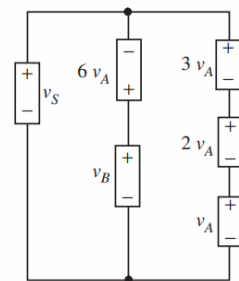
Highlight any circuit transformations used

Q. 1 In the given circuit, the two batteries are identical, each with an open-circuit voltage of 1.5 V. The lamp has a resistance of 5Ω when lit. With the switch closed, 2.5 V is measured across the lamp.

- Draw an abstracted equivalent circuit with ideal elements.
- What is the internal resistance of each battery?

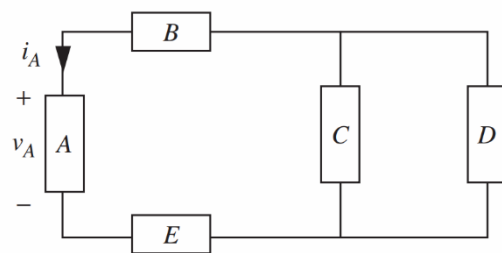


Q. 2 Determine the voltages v_A and v_B (in terms of v_S) for the network



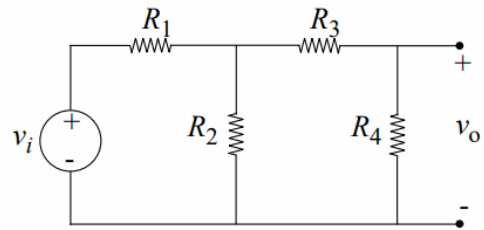
Q. 3 a) Assign branch voltages and branch current variables to each element in the network shown

- b) How many linearly independent KVL equations can be written for this network?
- c) How many linearly independent KCL equations can be written for this network?
- d) Formulate a set of KVL and KCL equations for the network.

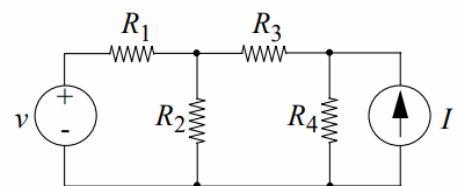


- e) Assign nonzero numbers to each branch current such that your KCL equations are satisfied.
- f) Assign non-zero numbers to each branch voltage such that your KVL equations are satisfied.

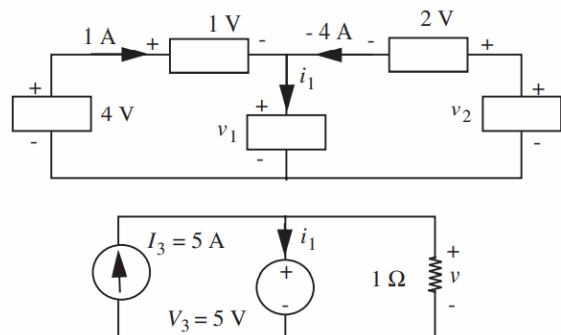
Q. 4 Design a resistor attenuator to make $v_o = v_i/1000$, using the circuit configuration given in figure.



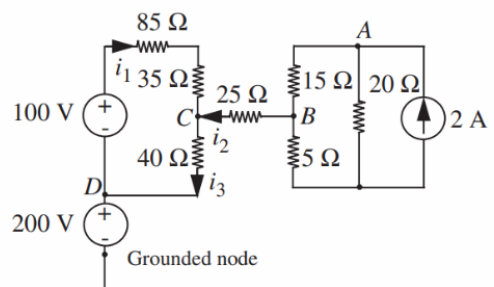
Q. 5 Solve for the voltage across resistor R_4 .



Q. 6 In each network in the given figure, find the numerical values of the indicated variables.



Q. 7 Find the voltage between node C and the grounded node



Q. 8 Find the potential difference between each of the lettered nodes (A, B, C, and D) and ground

